

```
1 using FinalExamCode.Models;
2 using JwtAuthDemo.DTOS;
3 using Microsoft.AspNetCore.Mvc;
4 using Microsoft.EntityFrameworkCore;
5 using Microsoft.IdentityModel.Tokens;
6 using System.IdentityModel.Tokens.Jwt;
7 using System.Security.Claims;
8 using System.Text;
9
10 namespace JwtAuthDemo.Controllers
11 {
12     [ApiController]
13     [Route("api/[controller]")]
14     public class AuthController : ControllerBase
15     {
16         private readonly IConfiguration _config;
17         private readonly AppDbContext _context;
18
19         public AuthController(IConfiguration config, AppDbContext context)
20         {
21             _config = config;
22             _context = context;
23         }
24
25         [HttpPost("login")]
26         public IActionResult Login([FromBody] LoginModel model)
27         {
28             // tìm user
29             var user = _context.Users
30                 .Include(x => x.UserRoles)
31                 .ThenInclude(x => x.Role)
32                 .FirstOrDefault(x => x.Username == model.Username);
33
34             if (user == null)
35             {
36                 return Unauthorized(new
37                 {
38                     message = "Sai tài khoản"
39                 });
40             }
41
42             // check password hash
43             bool checkPassword = BCrypt.Net.BCrypt.Verify(
44                 model.Password,
45                 user.Password
46             );
47
48             if (!checkPassword)
49             {
```

```
50         return Unauthorized(new
51         {
52             message = "Sai mật khẩu"
53         });
54     }
55
56     // lấy danh sách role
57     var roles = user.UserRoles
58         .Select(x => x.Role.Name)
59         .ToList();
60
61     // tạo token
62     var token = GenerateToken(
63         user.Id,
64         user.Username,
65         roles
66     );
67
68     return Ok(new
69     {
70         message = "Đăng nhập thành công",
71         token = token,
72         username = user.Username,
73         roles = roles
74     });
75 }
76
77 private string GenerateToken(
78     int userId,
79     string username,
80     List<string> roles
81 )
82 {
83     var claims = new List<Claim>
84     {
85         new Claim("UserId", userId.ToString()),
86         new Claim(ClaimTypes.Name, username),
87         new Claim(
88             JwtRegisteredClaimNames.Jti,
89             Guid.NewGuid().ToString()
90         )
91     };
92
93     // add roles vào token
94     foreach (var role in roles)
95     {
96         claims.Add(
97             new Claim(ClaimTypes.Role, role)
98         );
99     }
```

```
99         }
100
101         var key = new SymmetricSecurityKey(
102             Encoding.UTF8.GetBytes(
103                 _config["Jwt:Key"]
104             )
105         );
106
107         var creds = new SigningCredentials(
108             key,
109             SecurityAlgorithms.HmacSha256
110         );
111
112         var token = new JwtSecurityToken(
113             issuer: _config["Jwt:Issuer"],
114             audience: _config["Jwt:Audience"],
115             claims: claims,
116             expires: DateTime.Now.AddHours(1),
117             signingCredentials: creds
118         );
119
120         return new JwtSecurityTokenHandler()
121             .WriteToken(token);
122     }
123 }
124 }
125
126
127
128 =====
129 using FinalExamCode.Models;
130 using Microsoft.AspNetCore.Http;
131 using Microsoft.AspNetCore.Mvc;
132 using Microsoft.EntityFrameworkCore;
133
134 namespace FinalExamCode.Controllers
135 {
136     [Route("api/[controller]")]
137     [ApiController]
138     public class BookApiController : ControllerBase
139     {
140         private readonly AppDbContext _context;
141
142         public BookApiController(AppDbContext context)
143         {
144             _context = context;
145         }
146
147         // Lấy danh sách sách
```

```
148 [HttpGet("ListBooks")]
149 public IActionResult Index()
150 {
151     var books = _context.Book.ToList();
152     return Ok(books);
153 }
154
155 // Hiển thị sách theo ID
156 [HttpGet("GetBookById/{id}")]
157 public IActionResult GetBookById(int id)
158 {
159     var book = _context.Book.Find(id);
160
161     if (book == null)
162     {
163         return NotFound("Khum tìm thấy sách với ID này ời bé ơi");
164     }
165
166     return Ok(book);
167 }
168
169 // Thêm sách mới
170 [HttpPost("CreateBook")]
171 public IActionResult CreateBook(Book book)
172 {
173     _context.Book.Add(book);
174     _context.SaveChanges();
175
176     return Ok(book);
177 }
178
179 // Sửa sách
180 [HttpPut("UpdateBook/{id}")]
181 public IActionResult UpdateBook(int id, Book updatedBook)
182 {
183     var book = _context.Book.Find(id);
184
185     if (book == null)
186     {
187         return NotFound("Không tìm thấy sách");
188     }
189
190     book.Title = updatedBook.Title;
191     book.Author = updatedBook.Author;
192     book.CoverUrl = updatedBook.CoverUrl;
193     book.CategoryId = updatedBook.CategoryId;
194     book.PdfUrl = updatedBook.PdfUrl;
195
196     _context.SaveChanges();
```

```
197
198         return Ok(book);
199     }
200
201     // Xóa sách
202     [HttpDelete("DeleteBook/{id}")]
203     public IActionResult DeleteBook(int id)
204     {
205         var book = _context.Book.Find(id);
206
207         if (book == null)
208         {
209             return NotFound("Không tìm thấy sách");
210         }
211
212         _context.Book.Remove(book);
213         _context.SaveChanges();
214
215         return Ok("Xóa thành công");
216     }
217 }
218 }
219
220 =====
221 using FinalExamCode.Models;
222 using JwtAuthDemo.DTOs;
223 using Microsoft.AspNetCore.Mvc;
224 using Microsoft.EntityFrameworkCore;
225
226 namespace JwtAuthDemo.Controllers
227 {
228     [ApiController]
229     [Route("api/[controller]")]
230     public class UserController : ControllerBase
231     {
232         private readonly AppDbContext _context;
233
234         public UserController(AppDbContext context)
235         {
236             _context = context;
237         }
238
239         // Thêm mới user
240         [HttpPost]
241         public IActionResult Create(RegisterModel model)
242         {
243             bool exists = _context.Users
244                 .Any(x => x.Username == model.Username);
245         }
246     }
247 }
```

```
246         if (exists)
247         {
248             return BadRequest("Username đã tồn tại");
249         }
250
251         var user = new Users
252         {
253             Username = model.Username,
254             Password = BCrypt.Net.BCrypt.HashPassword(model.Password)
255         };
256
257         _context.Users.Add(user);
258         _context.SaveChanges();
259
260         foreach (var roleId in model.RoleIds)
261         {
262             _context.UserRole.Add(new UserRole
263             {
264                 UserId = user.Id,
265                 RoleId = roleId
266             });
267         }
268
269         _context.SaveChanges();
270
271         return Ok("Tạo tài khoản thành công");
272     }
273
274     // Sửa user
275     [HttpPut("{id}")]
276     public IActionResult Update(int id, UpdateUserModel model)
277     {
278         var user = _context.Users
279             .Include(x => x.UserRoles)
280             .FirstOrDefault(x => x.Id == id);
281
282         if (user == null)
283         {
284             return NotFound();
285         }
286
287         user.Username = model.Username;
288
289         if (!string.IsNullOrEmpty(model.Password))
290         {
291             user.Password = BCrypt.Net.BCrypt.HashPassword
292                 (model.Password);
293         }
```

```
294         // Xóa role cũ
295         _context.UserRole.RemoveRange(user.UserRoles);
296
297         // Add role mới
298         foreach (var roleId in model.RoleIds)
299         {
300             _context.UserRole.Add(new UserRole
301             {
302                 UserId = user.Id,
303                 RoleId = roleId
304             });
305         }
306
307         _context.SaveChanges();
308
309         return Ok("Cập nhật thành công");
310     }
311
312     // Xóa user
313     [HttpDelete("{id}")]
314     public IActionResult Delete(int id)
315     {
316         var user = _context.Users
317             .Include(x => x.UserRoles)
318             .FirstOrDefault(x => x.Id == id);
319
320         if (user == null)
321         {
322             return NotFound();
323         }
324
325         _context.UserRole.RemoveRange(user.UserRoles);
326
327         _context.Users.Remove(user);
328
329         _context.SaveChanges();
330
331         return Ok("Xóa thành công");
332     }
333
334     // Danh sách user
335     [HttpGet]
336     public IActionResult GetAll()
337     {
338         var users = _context.Users
339             .Include(x => x.UserRoles)
340             .ThenInclude(x => x.Role)
341             .Select(x => new
342             {
```

```
343             x.Id,  
344             x.Username,  
345             Roles = x.UserRoles  
346                 .Select(r => r.Role.Name)  
347             })  
348             .ToList();  
349  
350         return Ok(users);  
351     }  
352 }  
353 }
```